

Chapter I

Macrofinancial Risks

The global financial system remains resilient and financial conditions stable despite an uncertain and challenging backdrop marked by last mile disinflation process, elevated public debt, geopolitical tensions and economic fragmentation. The Indian economy and the financial system, on the other hand, strengthened by solid macroeconomic fundamentals and healthy balance sheets of financial institutions, are exhibiting robust growth and greater resilience.

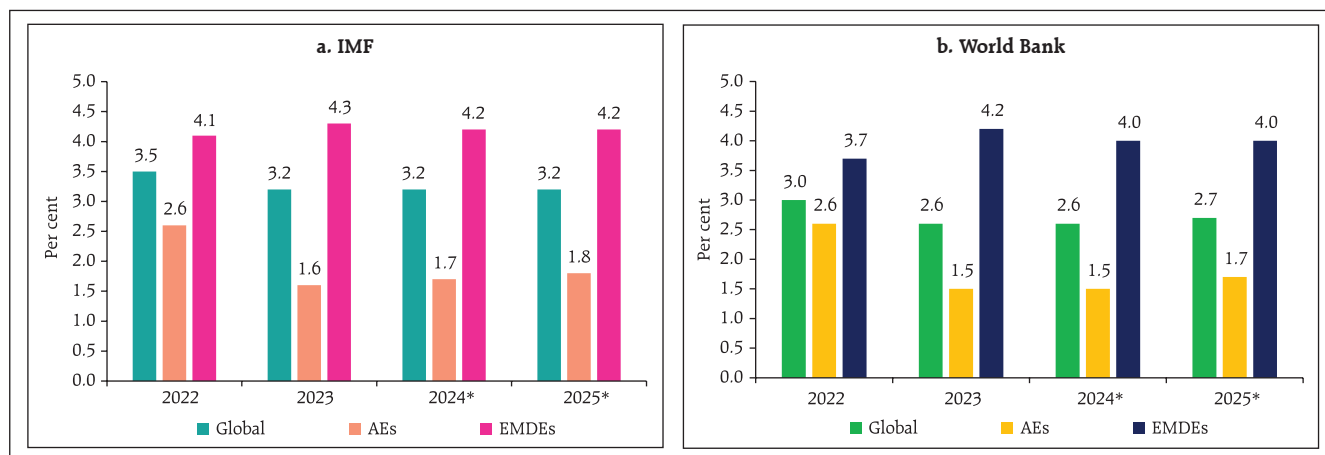
Introduction

1.1 Since the December 2023 issue of the Financial Stability Report (FSR), the global financial system has shown remarkable resilience, weathering the halting progress in the last mile of disinflation, financial stability risks stirred up by higher for longer stance of monetary policy and bouts of volatility unsettling financial markets on incoming data. As market expectations about the future course of monetary policy re-aligned with policy guidance, financial conditions have stabilised and risks of a hard landing have receded. Nonetheless, financial stability risks remain significant in the context of diverging economic conditions across geographies, stretched asset

valuations amidst flux in volatility fear gauges, high levels of global debt and commercial real estate strains. Prolonged geopolitical tensions and potential shocks to logistics and supply chains remain a clear and present threat with spillovers to commodity prices, the ongoing disinflation and eventually to the global economic outlook.

1.2 Against this backdrop, the International Monetary Fund (IMF) has projected¹ global growth to remain steady at 3.2 per cent in 2024 as in 2023, 0.3 percentage points higher than in its October 2023 World Economic Outlook update (Chart 1.1 a). The World Bank, on the other hand, projects global growth to be lower at 2.6 per cent in 2024 (Chart 1.1 b).

Chart 1.1: Global Growth Forecast

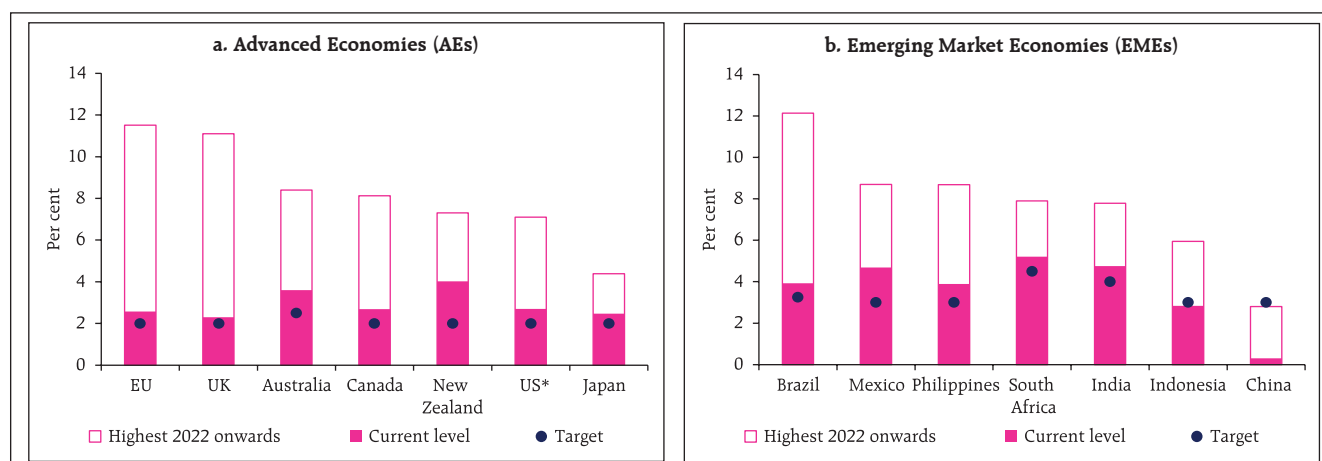


Note: * Forecasts.

Sources: IMF & World Bank.

¹ International Monetary Fund (2024), "World Economic Outlook - Steady but Slow: Resilience amid Divergence", April.

Chart 1.2: Global Headline Inflation



Note: (1) * Personal Consumption Expenditures Price Index for the U.S., and Consumer Price Index for all other countries.
 (2) Latest available data as on June 14, 2024.

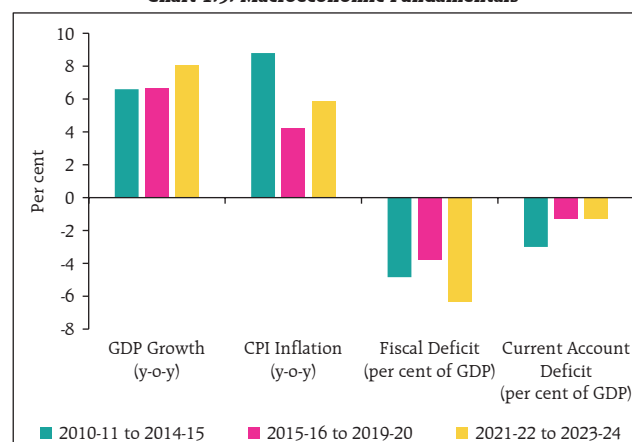
Source: Bloomberg.

1.3 The goal of bringing inflation down to targets remains the key focus of monetary policy authorities the world over, although headwinds are being encountered from sticky services inflation, elections across half the world's population, and the recent firming up of commodity prices, besides persisting geopolitical tensions (Chart 1.2 a and b).

1.4 In this uncertain international economic and financial environment, the Indian economy is exhibiting resilience and remains the fastest growing major economy. Moreover, India's contribution to global growth is rising and currently stands at 18.5 per cent in 2023-24².

1.5 Real GDP growth is on a rising trajectory, supported by macroeconomic and financial stability. Inflation is moderating, *albeit* at an uneven pace, fiscal consolidation is underway and the external sector position is improving in spite of global spillovers. The domestic financial system is well buffered with strong capital and liquidity ratios, declining levels of asset impairment and rising profitability (Charts 1.3, 1.4, 1.5 a, b and c).

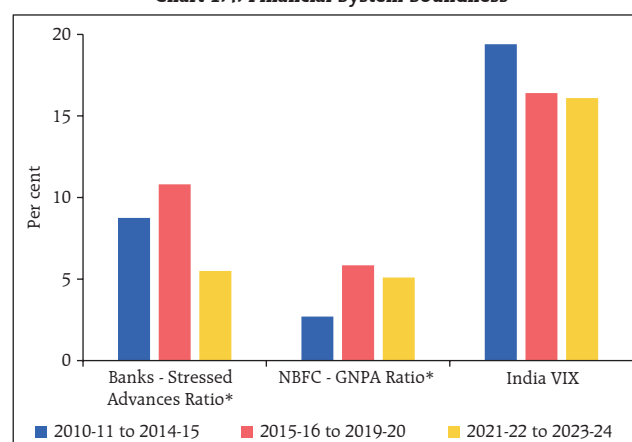
Chart 1.3: Macroeconomic Fundamentals



Note: FY 2020-21 (pandemic) not included.

Source: RBI.

Chart 1.4: Financial System Soundness



Note: (1) * Average ratio computed based on end-March figures of respective financial year.

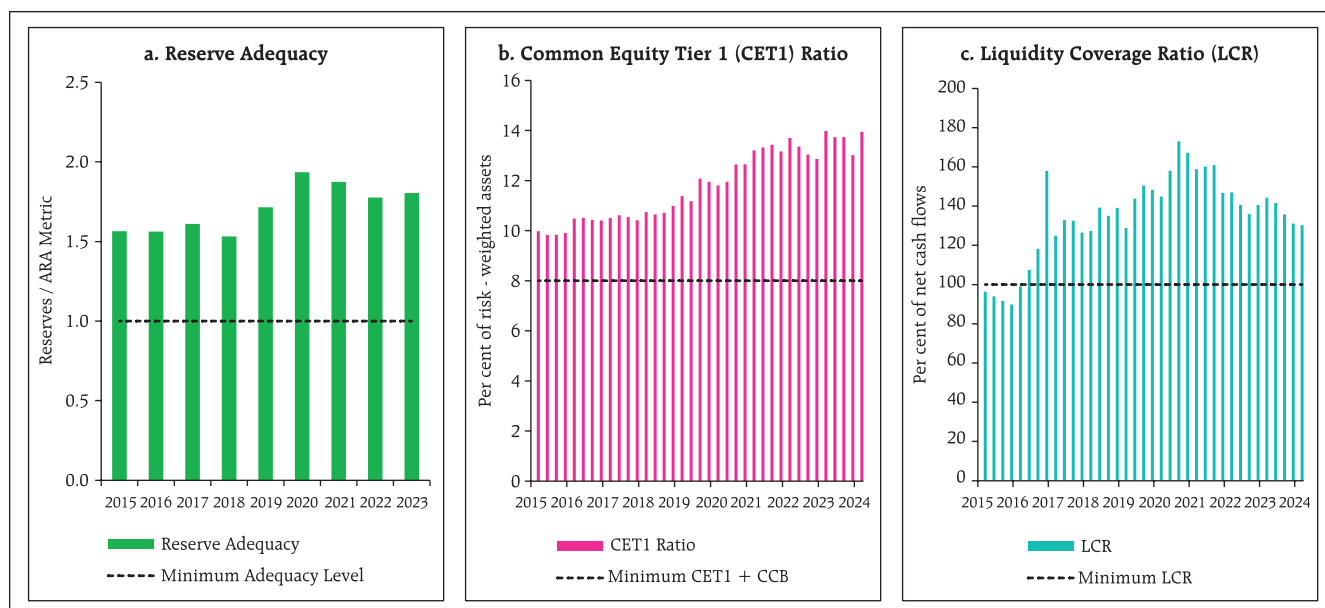
(2) FY 2020-21 (pandemic) not included.

(3) Stressed advances include gross non-performing assets and restructured standard advances.

Sources: RBI supervisory returns and Bloomberg.

² India's contribution to world GDP growth is calculated using the IMF World Economic Outlook database wherein GDP growth rate of countries is weighted by their share in world GDP based on purchasing power parity (PPP).

Chart 1.5: Reserve Adequacy and Banking System Buffers



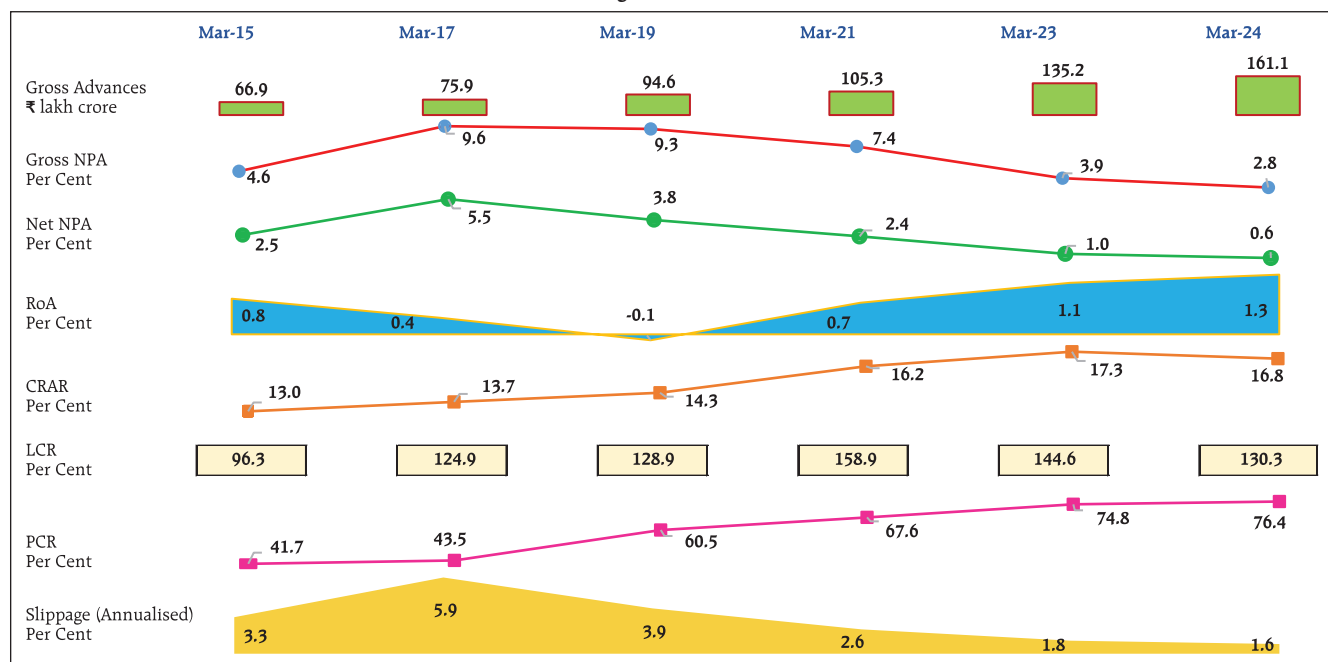
Note: Reserve Adequacy is based on the IMF's Assessing Reserve Adequacy (ARA) metric. CCB- Capital conservation buffer.

Sources: IMF and RBI.

1.6 India's financial system is bank dominated and, therefore, sound health of the banking system is a *sine qua non* for preserving financial stability. Banks' balance sheets are consistently improving, with multi-year low non-performing asset (NPA)

ratios, higher provisioning, stronger capital positions and robust earnings (Chart 1.6). In turn, these developments are catalysing a broad-based and sustained credit expansion.

Chart 1.6: Banking Sector Soundness Indicators



Note: As on June 14, 2024.

Source: RBI supervisory returns and staff calculations.

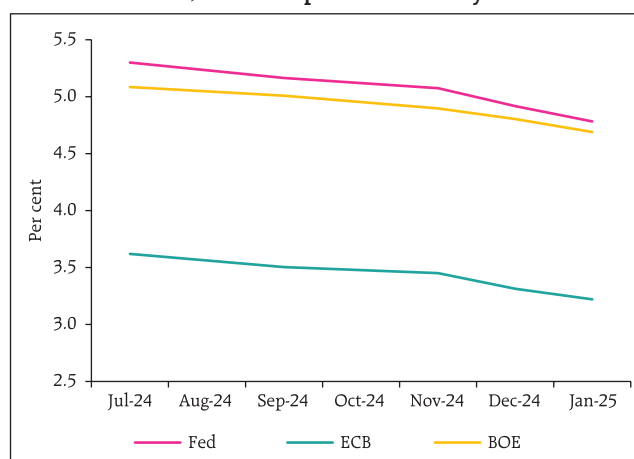
I.1 Global Backdrop

I.1.1. Macrofinancial Development and Outlook

1.7 A key feature of the evolving global economic outlook has been the recent alignment between market expectations and policy stances. The resilience of growth has allayed fears of hard landing and pushed out expectations of early interest rate cuts in response to central banks' discomfort about inflation remaining elevated relative to targets persistently; more recently, policy paths are increasingly expected to diverge as macroeconomic conditions move disparately in different regions of the world (Chart 1.7).

1.8 Financial conditions remain stable, with equity markets scaling new highs and credit spreads narrowing. Volatility has largely remained subdued but intermittent reversals have signalled unease in investor perceptions around potential flux (Chart 1.8 a and b).

Chart 1.7: Market Expectations of Policy Rates

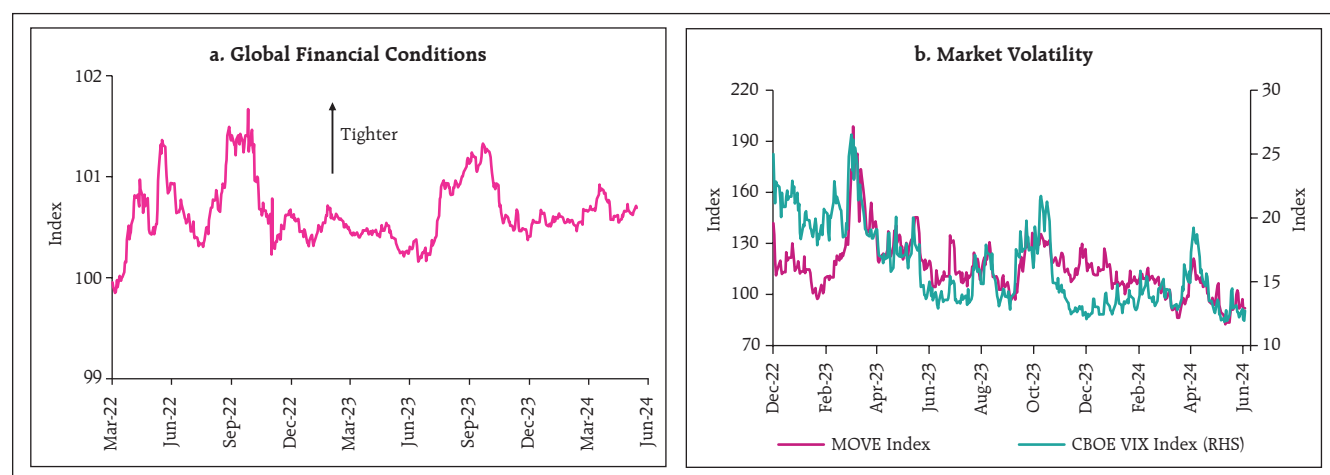


Note: Derived from overnight index swap (OIS) curve. As on June 14, 2024.

Source: Bloomberg.

1.9 The pace of balance sheet reduction of AE central banks has been measured so far even as they reassess banks' liquidity requirements in the context of ongoing quantitative tightening and fiscal developments (Chart 1.9).

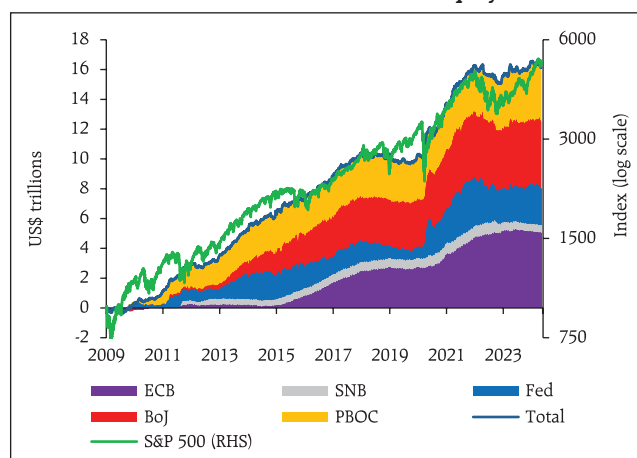
Chart 1.8: Financial Conditions and Volatility



Note: (1) Financial Conditions Index (FCI) is a composite index of individual country FCIs, based on policy rate, riskless long-term bond yield, corporate credit spread, equity price variable and trade-weighted exchange rate.

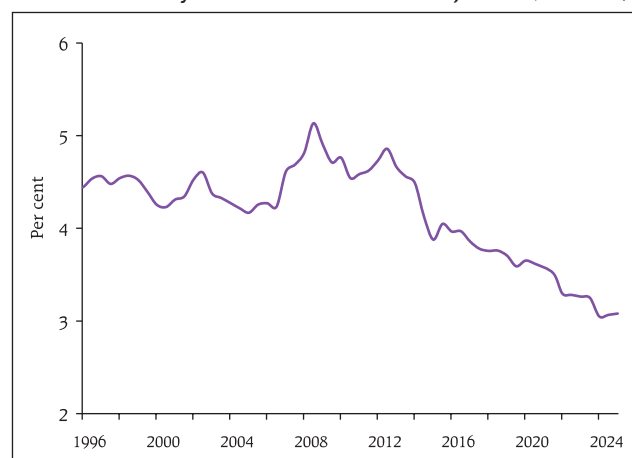
(2) CBOE - Chicago Board Options Exchange; MOVE - Merrill Lynch Option Volatility Estimate.

Sources: Goldman Sachs, Bloomberg.

Chart 1.9: Central Bank Reserves and Equity Prices

Note: Cumulative change in central bank reserves since 2009.

Sources: National central banks, S&P Global, Satori Insights.

Chart 1.10: Five-year-ahead Global Growth Projections (Real GDP)

Note: Date shows when forecast was made. Latest forecast is for 2029.

Source: IMF.

1.10 According to the IMF, the medium-term outlook³ is characterised by weak productivity and retreat in globalisation and global growth is expected to decline by more than a percentage point by the end of 2020s compared to the pre-pandemic average (Chart 1.10). Risks from geo-economic fragmentation and elevated debt call for closer global coordination of policy actions in order to secure and preserve macroeconomic and financial stability.

1.1.2. Global Macrofinancial Risks

1.11 While several near-term risks have receded, the global financial system continues to face heightened uncertainty surrounding the outer-term outlook for both financial markets and financial institutions. These shadows are cast by elevated levels of public debt, stretched asset valuations, stress in the commercial real estate (CRE) sector,

risks in private credit and cyber risk in addition to the persistence of geopolitical risks.

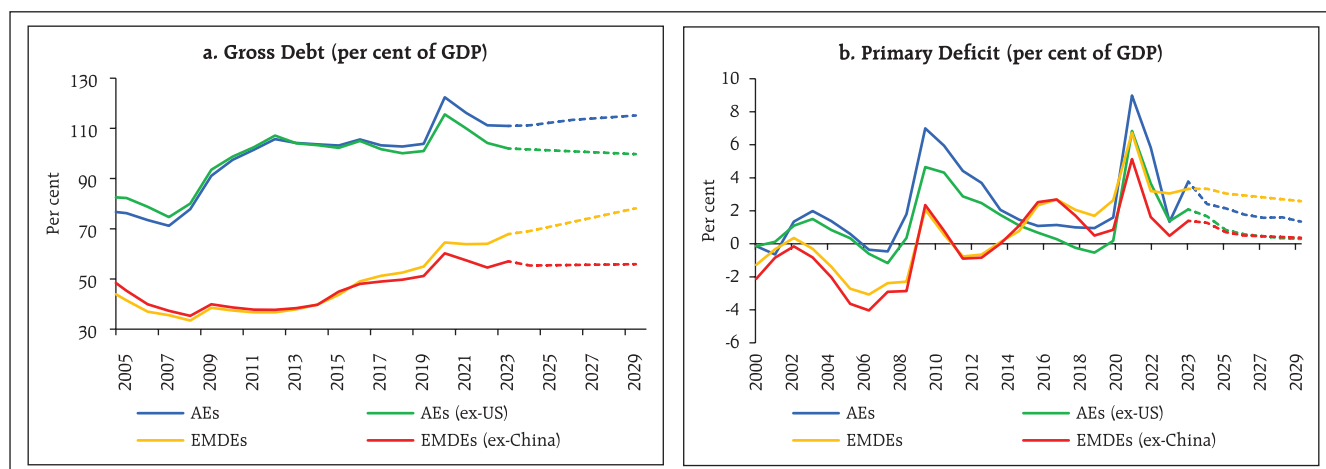
A. Elevated Public Debt

1.12 The frenetic expansion of global public debt in recent years has accentuated concerns about its sustainability. These worries are exacerbated by elevated interest rates and rollover risks for many overburdened economies. Global public debt has increased to 93.2 per cent of GDP by end-2023⁴ - nine percentage points above its pre-pandemic level – bloated by pandemic-era expansionary fiscal policies, post-pandemic supportive measures such as tax breaks and subsidies. The rollback of supportive measures has also been slow. Worryingly, the two largest economies in the world, *viz.*, the U.S. and China, are leading the increase in global debt (Chart 1.11 a and b).

³ International Monetary Fund (2024), "World Economic Outlook - Steady but Slow: Resilience amid Divergence", April.

⁴ International Monetary Fund (2024), "Fiscal Monitor: Fiscal Policy in the Great Election Year", April.

Chart 1.11: Global Public Debt



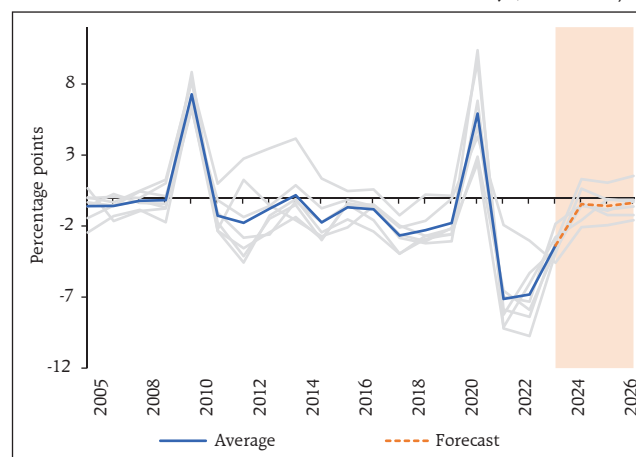
Note: Dotted lines represent forecasts.

Source: IMF.

1.13 The narrowing of the interest rate – economic growth rate differential against the backdrop of elevated interest rates and slowing growth is undermining debt sustainability (Chart 1.12). With several countries holding national elections in 2024, any related surge in government spending and widening of fiscal deficits may exacerbate the debt overhang, alongside costs associated with geopolitical tensions and green transition.

1.14 The threat of a looming debt crisis cannot be overemphasised. Fiscal consolidation took a backseat in 2023 as global fiscal deficit increased to 5.5 per cent of GDP on an average – a rise of 1.6 percentage points from the previous year. At the same time, high debt levels and narrowing interest-growth differentials have pushed debt-stabilising primary deficit lower by 2 percentage points of GDP on an average in advanced and emerging market

Chart 1.12: Interest Rate-Growth Rate Differential (G7 countries)



Note: (1) Forecast is based on real interest rates that are derived by deducting consumer price inflation from nominal 10-year government yields; Nominal yield forecasts are based on analyst estimates provided by Bloomberg while CPI and GDP growth forecasts are based on IMF estimates.

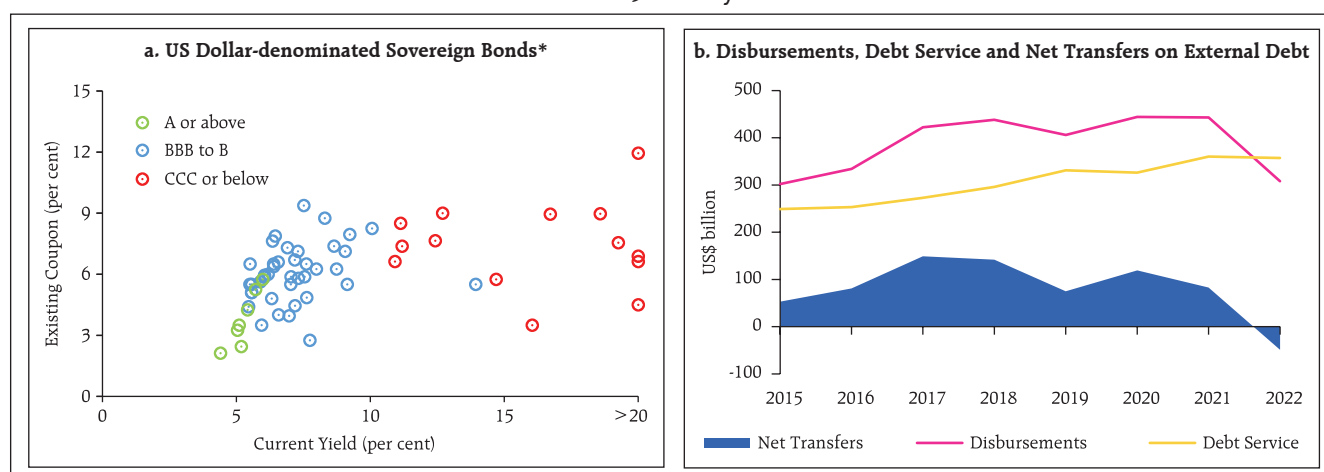
(2) Shaded region represents forecast.

Sources: IMF and Bloomberg.

economies⁵. Alongside mounting public debt, private debt also remains elevated. At US\$ 251.4 trillion, global debt stood at 239.9 per cent of GDP, with public debt at US\$ 97.7 trillion and private

⁵ International Monetary Fund (2024), "Fiscal Monitor: Fiscal Policy in the Great Election Year", April.

Chart 1.13: Debt Dynamics



Note: (1) * Each data point is an emerging market sovereign in the JP Morgan EMBIG Index. Rates pertain to the latest bond issuance by respective sovereign. Data as on May 31, 2024.

(2) Disbursements, debt service and net transfers on external public and publicly guaranteed debt of developing countries, based on a sample of 108 developing countries for which data are available. Net transfers correspond to disbursements minus debt service (principal and interest).

Sources: Bloomberg and UNCTAD

debt at US\$ 153.7 trillion as at end-2023⁶. We are in a debt supercycle with significant long-term structural imbalances⁷, which makes the financial system susceptible to frequent disruptions and overall macroeconomic instability.

1.15 The sharp rise in global public debt, especially among the two largest economies, poses significant spillover risks to EMEs, where low income and low-rated countries are particularly vulnerable. Surging interest rates have caused 18 sovereign defaults in 10 developing countries in the last three years alone, surpassing the total for the preceding two decades and interest payments have quadrupled in the past decade⁸. Meanwhile, according to the IMF, nine low-income countries are presently in debt distress, and an additional 25 countries are at high

risk of debt distress⁹. These economies face higher refinancing costs and elevated interest rates forcing them to make difficult choices in terms of servicing debt or making investments in health, education and infrastructure (Chart 1.13 a and b).

1.16 Coexistence of high debt levels and elevated interest rates can feed a vicious cycle of financial instability through impairment of government and private-sector balance sheets. For emerging and low-income countries, this poses challenges to their developmental prospects and their fight against hunger and disease, portending social unrest and conflict-related welfare losses. Global institutions and lenders need to consider measures to enable such borrowers to calibrate fiscal consolidation by considering country-specific circumstances.

⁶ Public debt and GDP are taken from IMF fiscal monitor and world economic outlook databases and private debt is sourced from the global debt monitor of the Institute of International Finance.

⁷ Mian, Atif (2024), "Breaking the Debt Supercycle", Finance and Development, IMF, March.

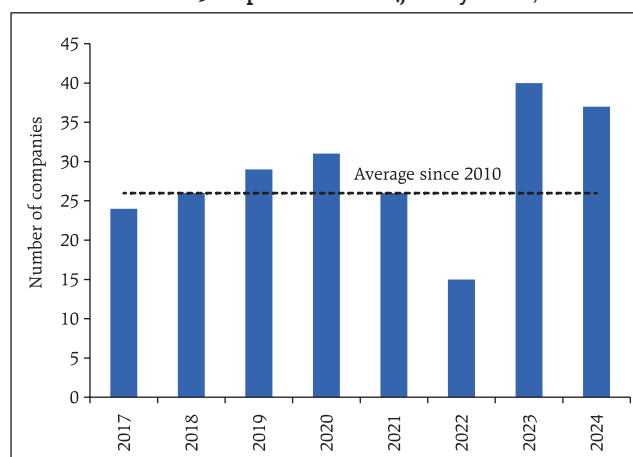
⁸ World Bank (2023), "International Debt Report", December.

⁹ IMF-World Bank Debt Sustainability Framework for Low Income Countries (risk ratings as on April 30, 2024).

B. Stretched Asset Valuations

1.17 Since the December 2023 FSR, financial markets have remained buoyant and generally risk-on. Equity markets and bond yields have rallied, volatility has declined and corporate bond spreads have narrowed (Chart 1.14 a, b and c). A sharp increase in prices of relatively riskier assets has followed in the wake of the exceptionally rapid monetary tightening. While stocks have been supported by strong earnings, with price-to-earnings (P/E) ratios close to historical averages, the narrowing of corporate spreads have coincided with rising episodes of corporate defaults (Chart 1.15). The average correlation among equities, bonds, credit and commodity indices is higher

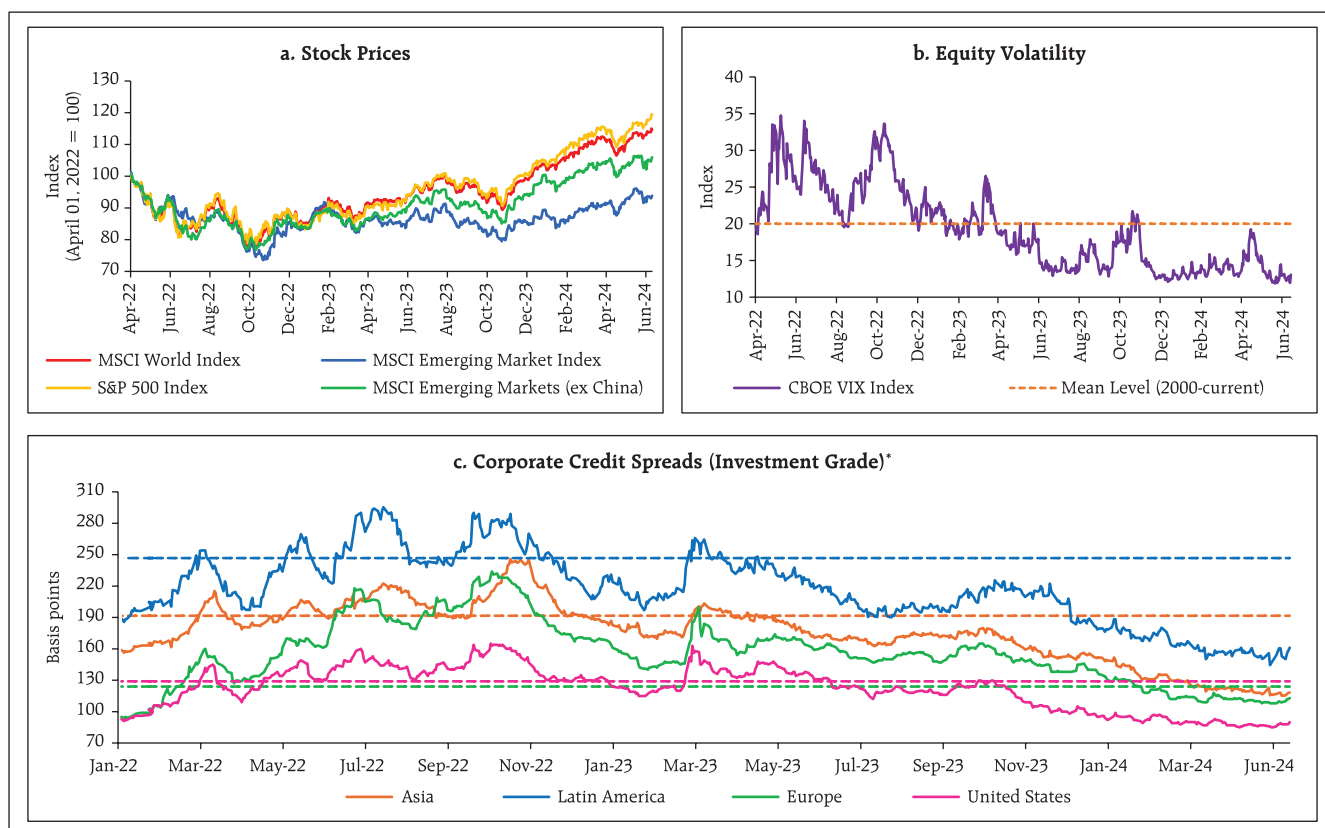
Chart 1.15 Corporate Defaults (January-March)



Source: S&P Global Ratings Credit Research & Insights

than the historical 90th percentile¹⁰ in both AEs and EMEs. The latest monetary policy cycle has witnessed a departure from past patterns when monetary tightening was associated with risk-off

Chart 1.14 Asset Valuations

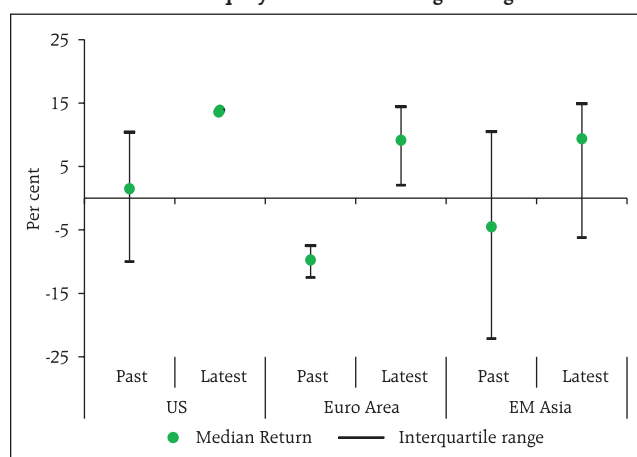


Note: * Dotted Lines indicate median value from 2005 to current date.

Sources: Bloomberg and JP Morgan.

¹⁰ International Monetary Fund (2024), "Global Financial Stability Report: The Last Mile: Financial Vulnerabilities and Risks", April.

Chart 1.16: Real Equity Returns in Post-tightening Plateaus



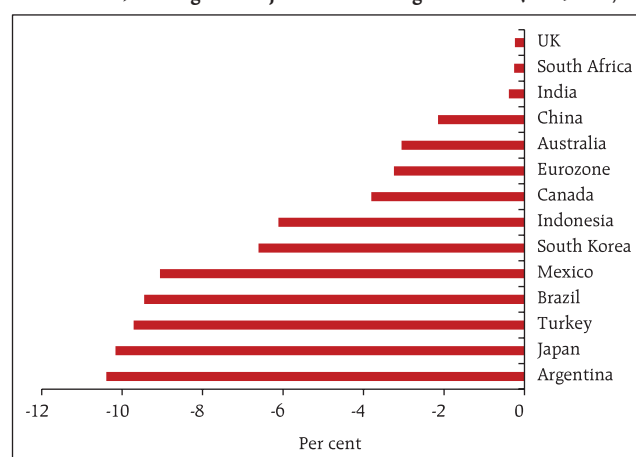
Note: Euro Area = France, Germany, Switzerland and Netherlands; EM Asia = China, India, Indonesia, South Korea. Monthly Data from January 1980 onwards; varying sample lengths across countries reflect data availability. Real equity returns between the latest hike in a tightening cycle and the first rate cut following it (or the end of time series if no such change)

Sources: BIS, FRED and Bloomberg.

sentiment and decline in prices of riskier assets (Chart 1.16). To the extent valuations are stretched, sudden shocks could precipitate stress that spreads contagiously across financial market segments through correlated sell-offs and band-wagon effects. Moreover, the growing importance of the role of non-bank institutions in financial intermediation and higher and hidden leverage could amplify stress even further in the face of large shocks, leading to materialisation of debt vulnerabilities and rise in credit losses.

1.18 Stretched asset valuations have been co-terminous with a strong US dollar (USD) as market expectations have gravitated towards prospects of higher interest rates in the US than in other economies. Moreover, the unrelenting strength of the USD is being bolstered by currencies such as Japanese Yen dropping to historic lows and sustained downward pressures on other currencies (Chart 1.17).

Chart 1.17: Change in Major Currencies against USD (2024 YTD)

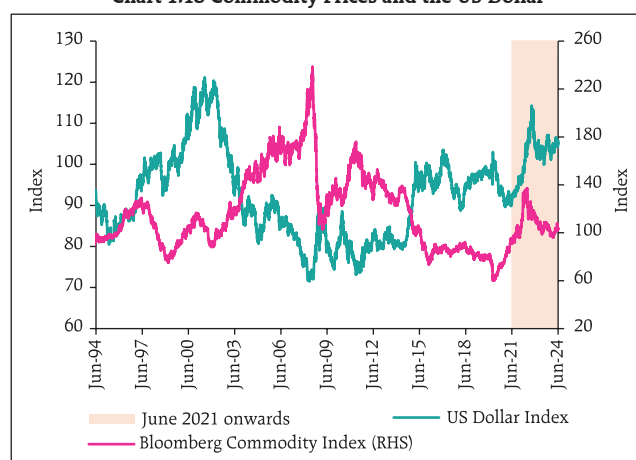


Note: As on June 14, 2024.

Source: Bloomberg

1.19 Besides interest rate differentials and currency depreciations, risks to the global economy, especially the EMEs, are being transmitted through multiple channels. First, commodity prices and the USD are moving in tandem in a break from their historical inverse behaviour¹¹ (Chart 1.18). This can compound inflationary pressures for EMEs, especially commodity importers, dampening

Chart 1.18 Commodity Prices and the US Dollar



Source: Bloomberg

¹¹ Rees, Daniel (2023), "Commodity prices and the US dollar", BIS Working Papers No 1083, March.

consumer spending, investment and economic growth. Second, the strong USD can increase debt service costs and prompt rating downgrades, limiting their access to global capital markets. Third, capital flows could become even more volatile.

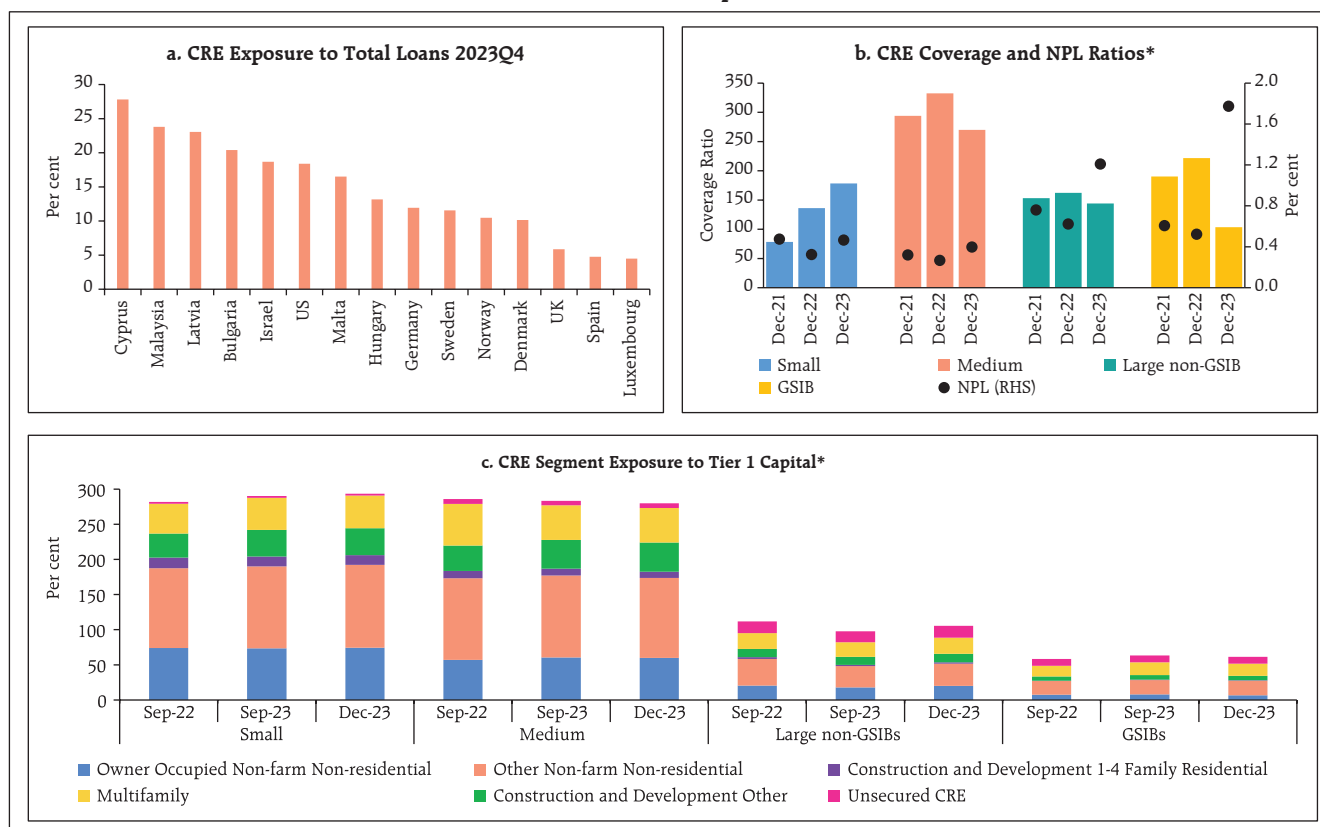
C. Stress in the Commercial Real Estate Sector

1.20 Commercial real estate (CRE) prices have declined sharply across countries. According to the IMF, CRE prices fell by 12 per cent globally in real terms over the past year¹² with the sector also remaining vulnerable to higher vacancy rates

and rising financing costs. CRE loans account for a substantial share of all bank loans in many countries and banks exhibit high sensitivity to expected and unexpected CRE losses, as reflected in relatively high CRE coverage ratios¹³ (Chart 1.19 a and b). Importantly, CRE exposures are concentrated in small and medium sized banks while large and global systemically important banks (GSIBs) have little exposure (Chart 1.19 c).

1.21 Increasingly, bank profitability and market valuations are interacting in multiple ways¹⁴.

Chart 1.19: CRE Exposures



Note: * Data based on select US banks. 'Small' refers to banks with less than US\$ 10 billion in total assets. 'Medium' corresponds to banks with assets between US\$ 10 billion and US\$ 100 billion. 'Large non-GSIB' corresponds to large banks with assets above US\$ 100 billion not classified as a GSIB and 'GSIB' corresponds to large banks classified as GSIBs; NPL = non-performing loans.

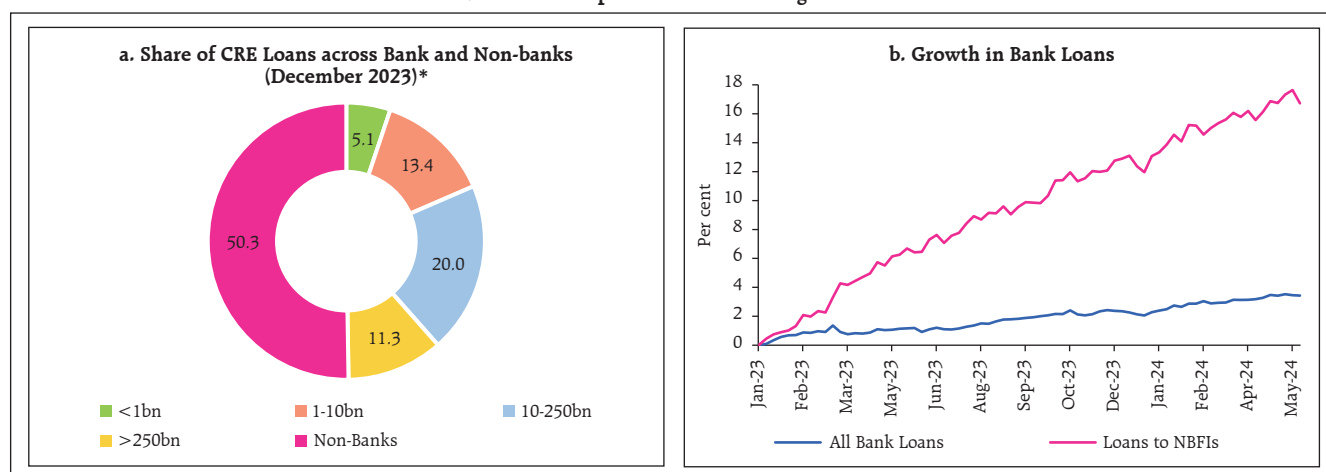
Source: IMF.

¹² International Monetary Fund (2024), "Global Financial Stability Report: The Last Mile: Financial Vulnerabilities and Risks", April

¹³ CRE coverage ratio is the ratio of loan-loss reserves to cover future losses to non-performing loans.

¹⁴ J Caparusso, U Lewrick and N Tarashev (2023), "Profitability, valuation and resilience of global banks - a tight link", BIS Working Paper, No 1144, November.

Chart 1.20 NBFIs – CRE Exposure and Interlinkages with Banks in US



Note: * Banks categorised based on asset size.

Sources: Federal Reserve and FDIC.

Short sellers have been targeting banks with large CRE exposure, as evident from the sharp fall in stock prices of a few banks that declared losses on their CRE portfolios. The resulting erosion of investor confidence makes it difficult for these banks to access funding markets which, in turn, undermines their resilience, especially if CRE prices decline further or outlier banks announce large losses. Another source of contagion stems from large exposure of non-bank financial institutions (NBFIs) to the CRE sector and their interlinkages with the banking system with potential knock-on effects on the broader banking system (Chart 1.20 a and b).

D. Geopolitical Risks

1.22 The recent flaring of military conflicts and their disruptive persistence has heightened the focus on risks to the global economy from geopolitical tensions, especially from their transmission through trade, inflation and confidence channels. Potential threats to financial

stability stem from volatility imparted to capital flows and asset prices, dampening of investor sentiment, unsettled financial markets and strains on balance sheets of financial institutions through deterioration in asset quality and profitability¹⁵.

1.23 The geopolitical risk index (GPRI)¹⁶ - constructed by counting the number of newspaper articles related to adverse geopolitical events as a share of the total number of newspaper articles at a monthly frequency - has spiked alongside sharp increases in trade restrictions and financial sanctions, reversing the gains from several decades of global economic integration. Inevitably, these developments portend strains to the international monetary system, undermining the efficiency of the global payment system (Chart 1.21 a and b).

1.24 Episodes of geopolitical tensions have generally been associated with spikes in crude oil prices and disruptions in supply conditions, which

¹⁵ European Central Bank (2024), "Financial Stability Review", May.

¹⁶ Caldara, Dario. and Iacoviello, Matteo (2022), "Measuring Geopolitical Risk", American Economic Review, Vol. 112, No 4, April, pp. 1194-1225.

Table 1.3: Sectoral Share in Sales of Companies with ICR < 1
(per cent)

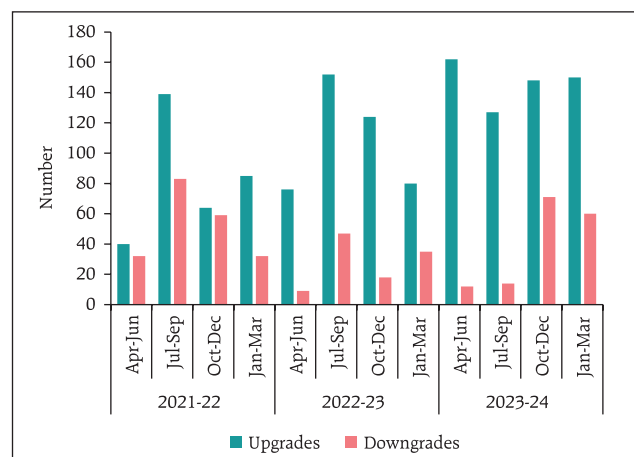
Year	Manufacturing	IT	Non-IT services	Aggregate
2020-21	6.8	3.0	44.1	10.5
2021-22	5.5	1.6	36.5	9.3
2022-23	3.5	1.3	22.5	5.5
2023-24	2.8	2.0	12.2	4.0

Source: Capitaline and RBI staff calculations.

1.46 The healthy credit profile of listed companies is also reflected in the rating actions of major credit rating agencies (CRAs). While credit upgrades have remained high in 2023-24 across three major credit rating agencies (ICRA, CRISIL and CARE), downgrades have also edged up moderately in H2:2023-24 (Chart 1.38).

1.2.5 Government Finance

1.47 Fiscal consolidation, buoyant tax collections and improvement in the quality of spending have been the distinguishing features of the Union Government's fiscal position. As per the provisional accounts (PA), the gross fiscal deficit

Chart 1.38: Trend in Credit Rating Actions

Sources: CRISIL, ICRA and CARE.

(GFD) was 5.6 per cent of GDP in 2023-24 as against the budget estimates (BE) of 5.9 per cent. Gross tax collections posted double digit growth, driven up by direct tax collections. On the expenditure side, the strategy remained geared towards growth-inducing capital expenditure; growth in revenue expenditure remained muted at 1.2 per cent (Table 1.4).

Table 1.4: Fiscal Indicators – Central Government

Fiscal Indicators	Amount (₹ crore)		Growth Rate (per cent)	
	2023-24(PA)	2024-25(BE)	2023-24(PA)	2024-25(BE)
1. Gross Tax Revenue	34,64,792	38,30,796	13.4	10.6
Direct Tax Revenue	19,56,645	21,98,830	17.9	12.4
Indirect Tax Revenue	15,08,147	16,31,966	8.2	8.2
2. Tax Revenue (Net)	23,26,524	26,01,574	10.9	11.8
3. Total Non-Debt Receipts	27,88,872	30,80,275	13.6	10.4
3. Total Expenditure	44,42,542	47,65,768	5.9	7.3
Revenue Expenditure	34,94,036	36,54,657	1.2	4.6
Capital Expenditure	9,48,506	11,11,111	28.2	17.1
4. Fiscal Deficit	16,53,670	16,85,494	-4.8	1.9
5. Revenue Deficit	7,65,624	6,53,383	-28.4	-14.7
6. Primary Deficit	5,89,799	4,95,054	-27.1	-16.1

Note: PA: Provisional Accounts; BE: Budget Estimate.

Sources: Union Budget Documents and Controller General of Accounts (CGA).